BACSE301 | EXPLORATORY DATA ANALYSIS | COURSE PROJECT

# Beyond GDP: What Really Makes a Country Livable?

*An exploratory data analysis of economic, health, social and environmental indicators for 146 countries (2023)*


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# Problem statement and objectives

GDP per capita is the usual yardstick for comparing countries, but it only measures output, not how people actually live. A rich country can still have pollution, weak healthcare or unequal schooling.

*"Is economic prosperity enough to make a country livable?"*

We use EDA to find patterns, not to prove causes.

## Objectives

1. See how GDP per capita relates to health and education outcomes
2. Check whether richer countries pay an environmental price ($CO_2$)
3. Measure correlations across the economic, health, social and environmental indicators
4. Find outlier countries that break the usual wealth pattern
5. Build a simple exploratory livability index and compare it with GDP rank

# Dataset overview

*One snapshot of 2023 country-level data, compiled from public sources*

**146** countries (rows)

**10** columns used, out of 35 (incl. country name)

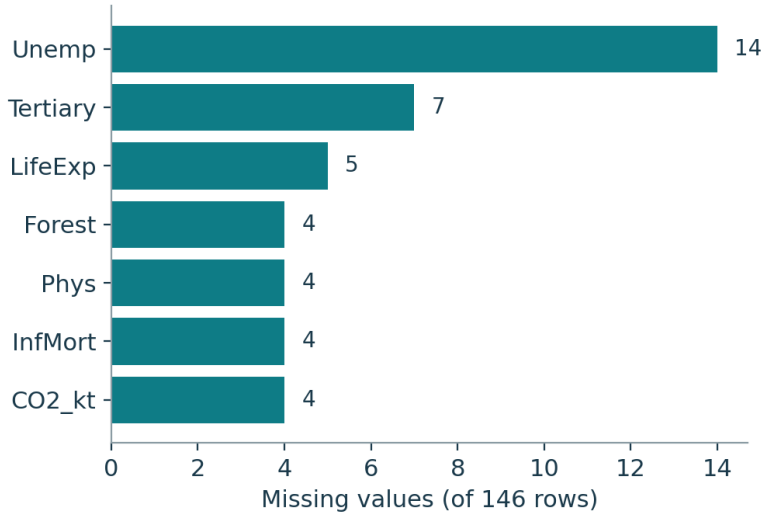
**2023** single-year snapshot

| Variable | Type | Dimension |
|---|---|---|
| GDP per capita (US$, derived) | Numeric | Economic |
| Unemployment rate (%) | Numeric | Economic |
| Life expectancy (years) | Numeric | Health |
| Infant mortality (per 1,000) | Numeric | Health |
| Physicians per 1,000 people | Numeric | Health |
| Tertiary enrolment (%) | Numeric | Social |
| $CO_2$ per capita (t, derived) | Numeric | Environmental |
| Forested area (% of land) | Numeric | Environmental |

Source: Global Country Information Dataset 2023 (Kaggle), built from World Bank and other public data. Our copy runs from Afghanistan to Saint Vincent alphabetically, so S–Z countries (USA, UK...) are missing.

# Data cleaning and preprocessing

*Mostly small fixes, but a few choices worth explaining*




Unemployment has the most gaps, so we avoided filling it with made-up values.

**1 Missing values**

Dropped 5 countries with no life expectancy (e.g. Andorra, Monaco, Nauru). Other gaps left as they are.

**2 Duplicates**

None. All 146 country names are unique.

**3 Inconsistent values**

Gross tertiary enrolment is above 100% for Australia, Greece and Grenada. Capped at 100.

**4 New columns**

GDP per capita = GDP / population. $CO_2$ per capita = $CO_2$ / population.

**5 Final table**

141 countries, 10 columns.

# Descriptive statistics

*Averages hide a lot: the median country is much poorer than the mean*

| Variable | Mean | Median | Std | Min | Max | Skew |
|---|---|---|---|---|---|---|
| GDP per capita ($) | 14,792 | 6,076 | 22,975 | 261 | 172,368 | 3.50 |
| Life expectancy | 72.3 | 73.9 | 7.5 | 52.8 | 84.2 | -0.58 |
| Infant mortality | 21.7 | 13.8 | 19.7 | 1.4 | 84.5 | 1.10 |
| Physicians /1,000 | 1.84 | 1.54 | 1.70 | 0.04 | 8.42 | 1.03 |
| Unemployment (%) | 6.67 | 5.20 | 4.93 | 0.09 | 23.41 | 1.34 |
| Tertiary enrol. (%) | 38.5 | 34.4 | 28.2 | 0.8 | 100 | 0.43 |
| $CO_2$ per capita (t) | 4.10 | 2.42 | 5.09 | 0.02 | 36.49 | 2.94 |
| Forest area (%) | 32.0 | 31.7 | 23.4 | 0 | 91.9 | 0.48 |

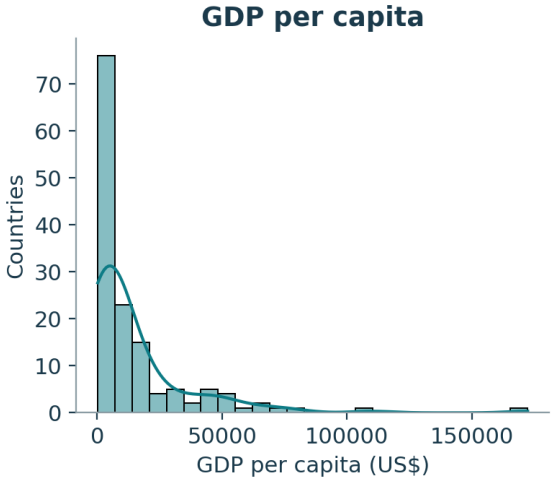
GDP per capita is very right-skewed (3.5): mean $14.8k, median $6.1k.

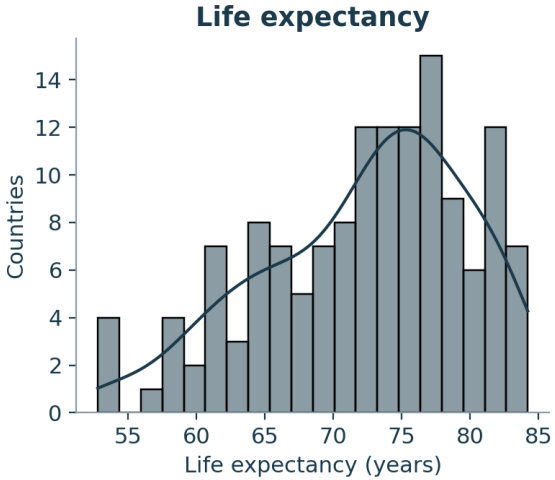
Life expectancy leans the other way, with a tail of countries in the 50s.

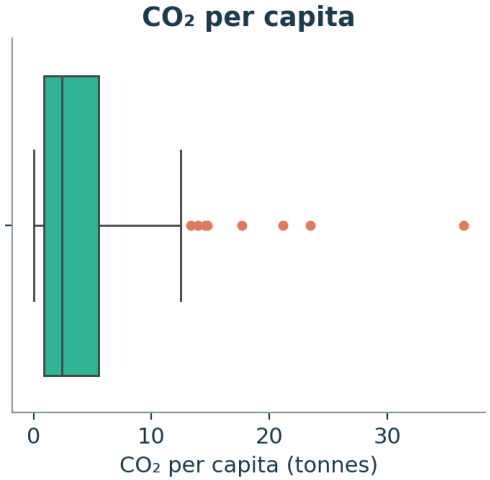
A few heavy emitters pull $CO_2$ per capita up: mean 4.1 t vs median 2.4 t.

# Univariate analysis

*Three variables, three very different shapes*




### GDP per capita

Long right tail. After a log transform the skew drops from 3.5 to about 0, so later plots use a log axis.

### Life expectancy

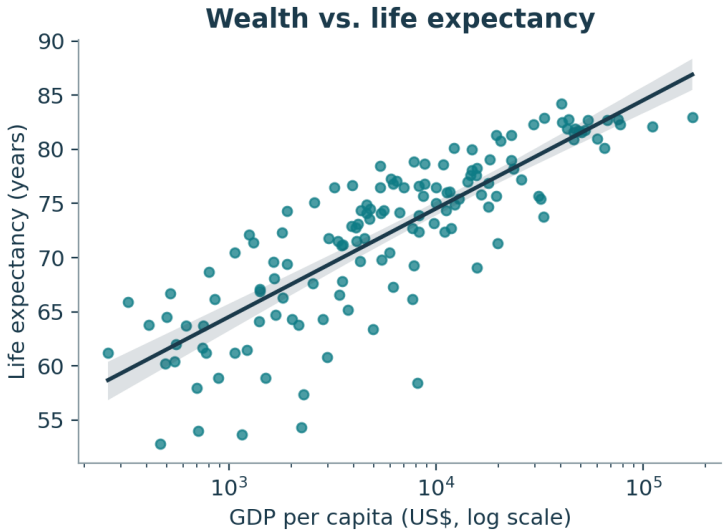
Most countries sit between 70 and 82 years, with a second small group in the 50s and 60s.

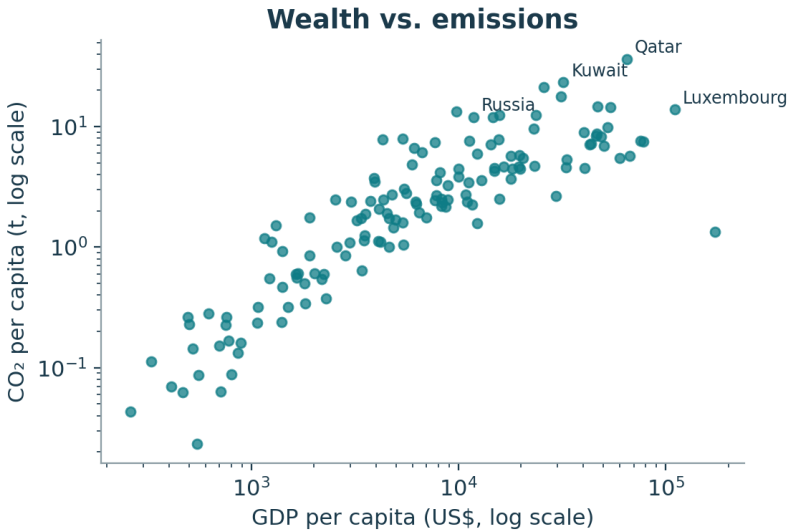
### $CO_2$ per capita

Median is 2.4 t. Eight countries sit beyond the 12.5 t boxplot whisker.

# Bivariate analysis

*Wealth buys years of life, and it also buys emissions*



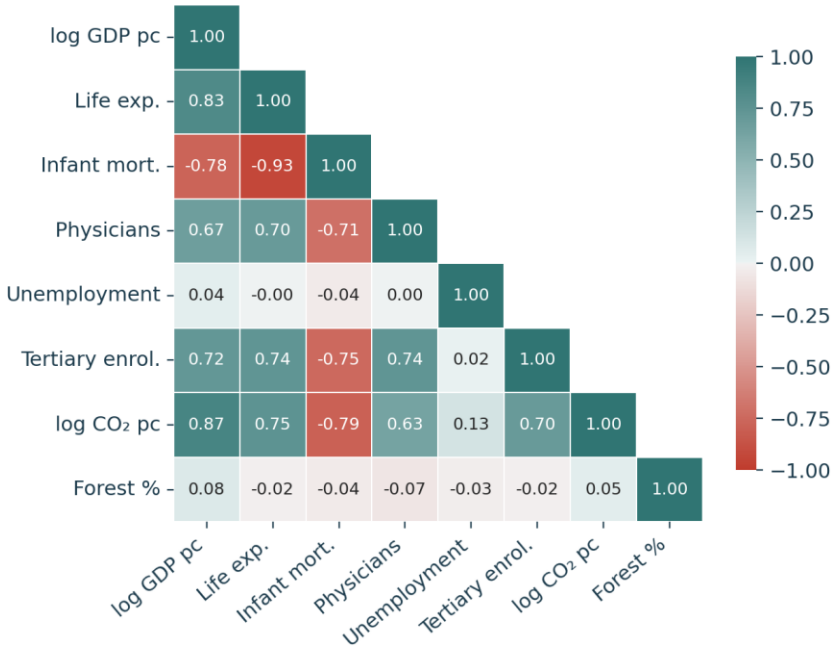




**Life expectancy:** r = 0.83 with log GDP. Each doubling of GDP per capita goes with about +3 years, and the gains flatten at the top.

**$CO_2$:** r = 0.87 on log-log axes. The Gulf states and Luxembourg sit far above the trend.

# Multivariate and correlation analysis

*Health variables move together; unemployment and forests do not*




**Strongest link:** infant mortality and life expectancy, r = −0.93.

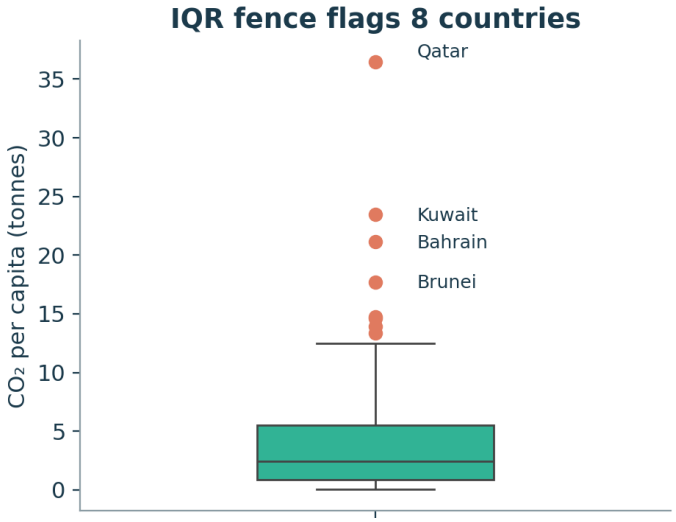
**Log GDP** goes with physicians (0.67), tertiary enrolment (0.72) and life expectancy (0.83).

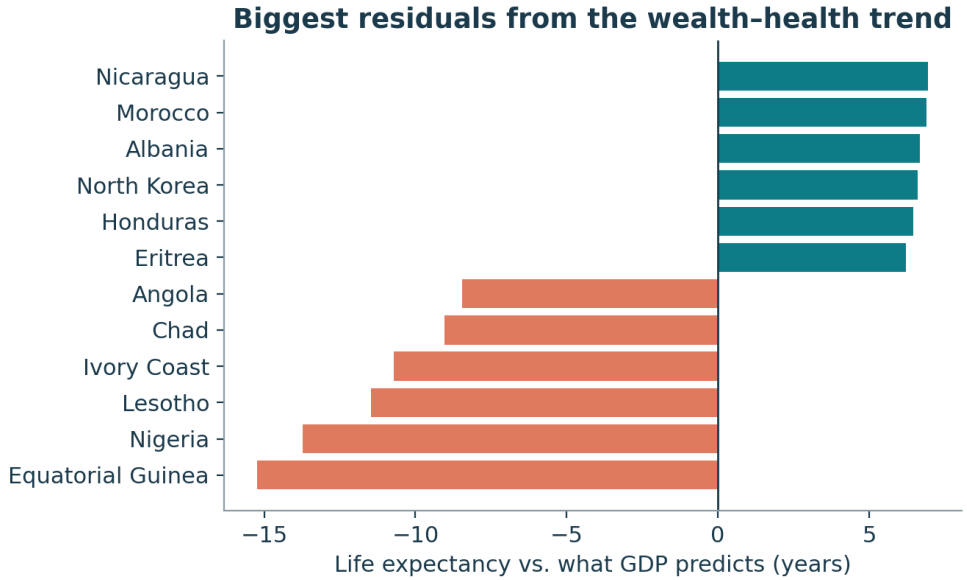
**$CO_2$ per capita** is positively linked with life expectancy (0.75), which is the trade-off in one number.

**Unemployment and forest cover** are close to zero against GDP (0.04 and 0.08). Money does not fix these.

# Outlier analysis

*Two ways to be unusual: emit too much, or live shorter than your income predicts*



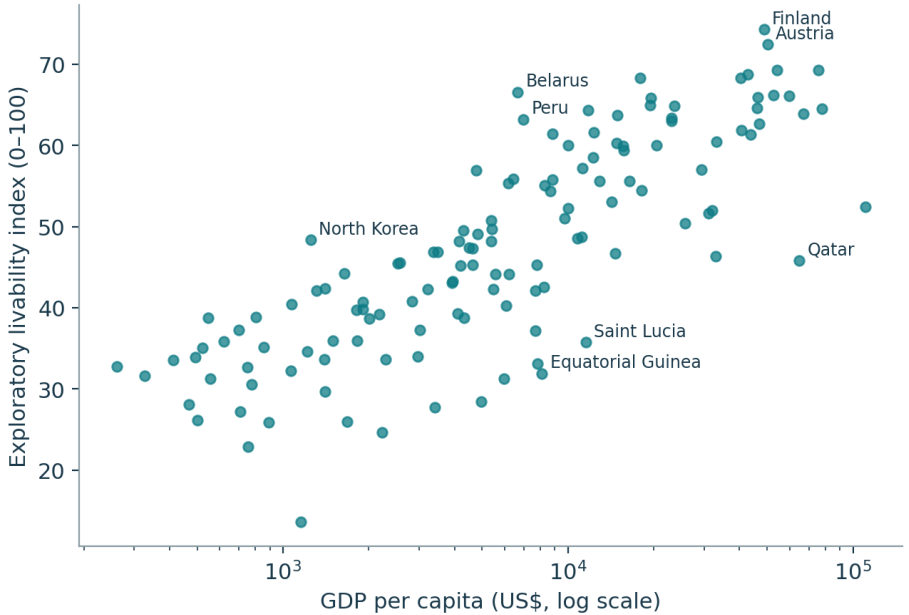




IQR rule: Qatar (36.5 t), Kuwait, Bahrain, Brunei, Canada, Australia, Luxembourg and Kazakhstan all pass the 12.5 t fence.

Trend residuals: Equatorial Guinea lives 15 years less than its GDP predicts (resource wealth, low spending on people). Nicaragua, Morocco and Albania are 7 years above.

# Exploratory livability index

*A rough composite, for visualization only*




## How it is built

Each indicator is min-max scaled, grouped into health, social, economic and environmental scores, then averaged with equal weights (131 countries with complete data).

## What it shows

Top 5: Finland, Austria, Australia, Norway, New Zealand.

Falls vs GDP rank: Qatar 5th to 68th, Equatorial Guinea 55th to 116th.

Climbs: Belarus 61st to 8th, Peru 60th to 21st.

*Rank correlation with GDP is 0.83, partly because GDP is one input. Weights are our own choice, not an official score.*

# Key findings and insights

## +3 yrs

Life expectancy rises about 3 years for every doubling of GDP per capita (r = 0.83).

## 64 vs 81

Median life expectancy in the poorest vs richest income quarter. Infant mortality: 45 vs 3 per 1,000.

## 27x

Median $CO_2$ per capita goes from 0.26 t to 7.1 t across income quarters.

## ~0

Unemployment (r = 0.04) and forest cover (0.08) barely follow GDP.

## -15 yrs

Equatorial Guinea lives 15 years shorter than its income predicts. Nicaragua is 7 above.

## 63 places

Qatar drops from 5th by GDP to 68th on the index. Belarus climbs 53.

# Conclusion and future scope

## Is prosperity enough?

No, but it is a strong starting point. GDP per capita explains roughly 69% of the variation in life expectancy, which leaves almost a third to other things. It does nothing for unemployment or forest cover, and richer countries pay for it in emissions.

## Limitations

One year only, so these are associations, not causes. 146 countries, with gaps in several columns. Index weights are arbitrary.

## Future scope

- Use multi-year data to see how countries change over time
- Add happiness scores and air quality (PM2.5)
- Cluster countries into livability groups
- Try other index weightings and test how stable the ranks are
- Build a Power BI dashboard for exploring profiles

# References


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